FLEET TEAM DIGEST

737MAX-FTD-27-25001

Issue Title : 737MAX Control Cable Relaxation

Airplane Model	ATA	Minor Model(s)
737MAX	2709-14	-10, -7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
	9E991

Originated Date	Last Revised Date	Created On
07/16/2025	07/16/2025	05/19/2025

Estimated Completion Date	Next Update Date
12/28/2028	09/04/2025

Description

Due to endurance yields of 1/8” BMS7-265 Grade B and Grade D control cable material being too low to support BCA Production and spares, Boeing is only able to procure Grade K control cable material.

Endurance testing for control cable material, as defined in BMS7-265, includes a 35,000-cycle severe bend test that requires a 1000 lb residual breaking strength. Grade B (1/8” BMS7-265) material began to experience endurance testing failures in November 2017 affecting production yield. Extensive BPSM (Boeing Problem Solving Model) efforts were conducted to determine the root cause of these failures. The primary root cause was attributed to the heat treat process variation at a third-tier supplier. Due to internal changes regarding the heat treat process, the supplier was unable to consistently produce acceptable material and did not have documentation on how the Grade B material was being produced historically. An alternate material which has a higher zinc content, now classified as Grade D, was able to meet the endurance requirement. As a result, Boeing incorporated Grade D into production.

In 2020, Grade D control cables also began to experience endurance testing failures. As a result, endurance yields of both Grade D and Grade B material became too low to support BCA production. To make use of the material in the supply chain that successfully passed endurance testing, Boeing qualified a new alternate material, Grade K, which allowed for an even higher zinc content than Grade D. Grade K was cut in to production at line 8652.

The root cause of elevated control cable relaxation rate has been shown to be a creep of the zinc coating on each individual carbon steel wire, which can allow the helically wound wires to move closer to the centerline of the strands of the control cable when subjected to sufficient tension. Movement of the wires closer to the centerline of cable strands causes the control cable to increase in length resulting in reduced tension. Over time, the cable tension stabilizes and further tension reduction is significantly reduced.

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In response to elevated Grade K and Grade D cable relaxation, Boeing issued service letters 737-SL-27-265-C (Grade D), 737-SL-32-201-D (Grade D) , and 737-SL-27-307 (Grade K), recommending operators re-tension the control cables 3-4 months after delivery or installation. This period allows for the initial relaxation of the cables to reach a steady state. Although not a requirement, the intent of the 3-4 month re-tensioning recommendation is for operators to perform a one-time adjustment to these cables which, if not performed, could result in a service disruption due to a pilot write up regarding the control system "feel". After this re-tensioning, Grade D and Grade K control cables will continue to relax at a rate that is similar to Grade B control cables. Any subsequent relaxation and corresponding re-tensioning of cables through the life of the aircraft should be considered normal, regardless of the control cable's zinc content.

Status

Boeing is working to return to Grade B control cables and has developed a schedule which will be tracked in the "Milestones" section of this FTD article.

After incorporating Grade K control cables, Boeing began receiving reports from operators regarding insufficient lower turnbuckle adjustment when adjusting the stabilizer STA and STB control cables when performing the retensioning recommendation outlined in 737-SL-27-307. Although this issue has been reported on airplanes delivered with Grade B and Grade D material, the frequency of occurrence has been much higher for Grade K material. Please refer to the "Stabilizer Turnbuckle Adjustment" attachment.

Similar to reports of relaxation affecting the stabilizer control cables, Boeing has also received reports of operators experiencing brake cable system performance issues due to LGB1 and LGB2 control cable relaxation.

Please refer to the interim action section for further exploration of both issues.

Interim Action

Stabilizer Turnbuckle Adjustment

Boeing has received reports from operators running out of adjustment on the stabilizer forward mechanism lower turnbuckle before the minimum AMM tension is reached. In response, Boeing shortened the STA and STB Grade K cables to provide additional adjustment for the stabilizer control cables in service. The updated cable lengths were implemented when Grade K was cut into production.

When implementing the shortened STA and STB Grade K cables, Boeing received two reports from operators running out of turnbuckle adjustment on new aircraft when performing the re-tensioning service letter. As a result, Boeing is further shortening the stabilizer cables to allow for additional adjustment. Please refer to the "Stabilizer Cable Length Reduction" attachment for both the initial and new cable length reduction values for production and spares.

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In addition to shortening the STA/STB cables, Boeing has updated the production requirements in the Boeing Factory rigging document. The previous document required a C dimension value of 11.00 - 15.25", but to allow for sufficient turnbuckle adjustment after 3-4 months of relaxation, the factory rigging document has been updated to include a C dimension value of 12.25-15.25 inches. Operators would still be given the 11.00 - 15.25” C dimension in the AMM.

NOTE: If operators run out of turnbuckle adjustment, it is acceptable to shorten and re-terminate the cable ends in accordance with SOPM 20-50-16 to allow for proper adjustment. Refer to AMM 27-09-14 “Control Cables Identification and Assembly Specifications” for additional information when re-terminating STA/STB cables.

Brake System Performance

In recent months, operators have reported issues with parking brakes due to landing gear (LGB1 and LGB2) control cable relaxation. Cable relaxation allows the brake metering valves to be in a position that results in increased internal leakage when the parking brake is set, thereby affecting the accumulator bleed down rate. Please refer to the “Brake System Performance” attachment. Although this issue can still occur with Grade B and Grade D control cables, the issue has occurred much sooner into service for Grade K cables.

As a result, Boeing is currently working to cut over to Grade D cables for brakes ahead of the overall Grade D incorporation date found in the “Milestones” section of this FTD article.

Final Action

Boeing’s long-term goal is to return the supply chain to a position of being able to reliably produce sufficient quantities of 1/8” BMS7-265 Grade B material for the fleet.

Boeing is currently working with the current control cable supplier to address the root causes for endurance testing failures and implement corrective actions. The most recent batch of trial Grade D material passed with 100% yield.

Boeing has also identified an additional supplier. Grade B material has been ordered for testing. The results of this testing will be updated in the "Milestones" section of this FTD article

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

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Stabilizer Cable Length

Stabilizer Reduced Length Cable Incorporation - ECD: July-2025

Overall Return to Grade D

Additional Grade D Test Results (Current Supplier) - ECD:

March-2026 Cutover to Grade D - ECD: May-2026

Overall Return to Grade B

Grade B Test Results (Alternate Supplier) - ECD: August-2025

Grade B material order (Current Supplier) - ECD: June 2026

Brake Cable Status

Brake Cable Mitigation Action - ECD: TBD

Related Categories

- Manufacturing Quality Issues : Issues related to airplane or component manufacturing quality

Attachments:

a) Stabilizer Turnbuckle Adjustment (STA/STB)
b) Affected Control Cables and Zinc Content Table
c) Brake System Performance (LGB1 and LGB2 Cables)
d) Stabilizer (STA/STB) Cable Length Reduction

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